FINAL PROJECT · AI / ML

# A scalable recommender is a system, not a single model.

Large-scale personalized recommendation with retrieval, ranking, serving, experimentation, and orchestration.

DATASET
**25.0M ratings**

ONLINE DESIGN
**2 stages**

ORCHESTRATION
**10 tasks**

DATA CONTRACT

# 25 million ratings become a leakage-safe temporal split.

| TRAIN · ≤ cutoff | VALIDATION | TEST |
| --- | --- | --- |
| **19,986,521** | **2,501,547** | **2,512,027** |
| 79.95% · features learned here | 10.01% · model selection | 10.05% · final evaluation |

Feature cutoff · 22 June 2016 05:34:47 UTC



README · DATA_CONTRACT.md

SYSTEM DESIGN

# Two stages separate broad candidate recall from precise ranking.

| 01 | 02 | 03 | 04 | 05 | 06 |
|---|---|---|---|---|---|
| **Spark** | **Features** | **Two tower** | **FAISS** | **DCN / W&D** | **FastAPI** |
| clean + temporal split | users, items, genres | 64-D embeddings | top 200 candidates | contextual re-rank | cache + feedback |

**AIRFLOW** orchestrates existing reproducible commands and validates every artifact boundary

— RETRIEVAL

# A 64-dimensional user vector reduces 24,728 items to 200 candidates.

KNOWN USERS

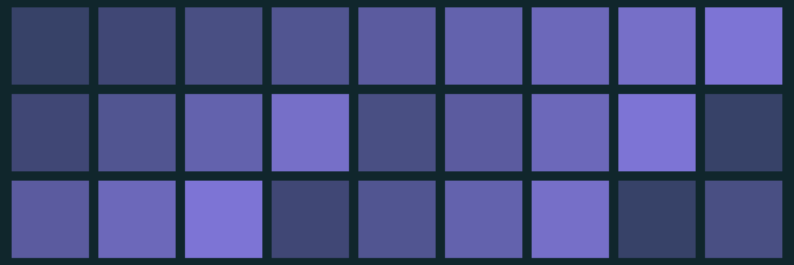
**137,622**

Coverage-preserving training vocabulary

→

EXPORTED ITEM MATRIX

**24,728 × 64**



→

RANKER INPUT

**Top 200**

Inner-product similarity, seen-item filtering



RETRIEVAL_CONTRACT.md

— VECTOR SEARCH

# At this catalog size, high-recall HNSW is roughly tied with exact search.

| INDEX | RECALL@200 | P50 MS | P95 MS | QUERIES/S |
|---|---|---|---|---|
| **Exact Flat** | 1.0000 | 0.2932 | 0.3196 | 3,622 |
| **HNSW · ef128** | 0.9402 | 0.1824 | 0.2230 | 5,868 |
| **HNSW · ef256** | 0.9859 | 0.2839 | 0.3307 | 3,680 |

500 seeded queries · K=200 · one CPU thread

**Measured trade-off, not hidden**



faiss_metrics.json

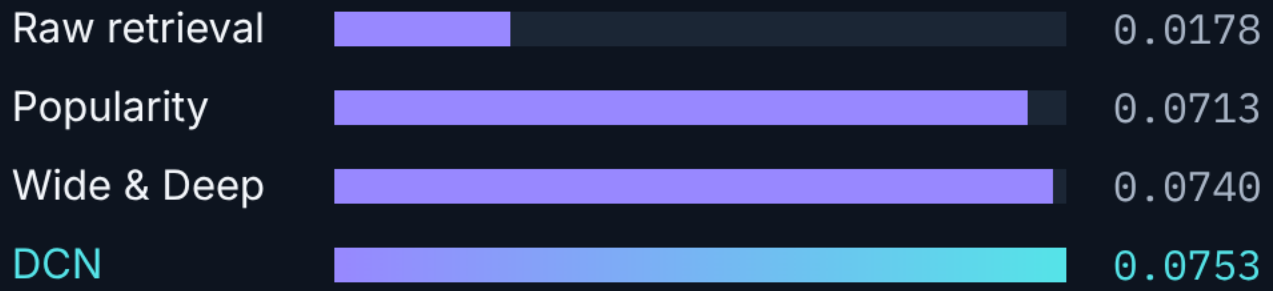
— DEEP RANKING

# DCN wins the ranking comparison by a small, reproducible margin.

## Validation NDCG@10

same 200 candidates · 500 users

**0.0753**




SELECTION RULE

## Future-validation NDCG@10

Early stopping restores the best validation-AUC epoch before SavedModel export.



ranking_metrics.json

ONLINE SERVING

# Serving stays available when the cache is missing or unavailable.

**01 · Request**

Known user or cold-start session profile

→

**02 · Compute**

Tower → FAISS → assigned ranker

→

**03 · Respond**

Versioned result + impression event

| FIRST MISS | WARM P50 | WARM P95 |
|---|---|---|
| 374.10 ms | 13.79 ms | 15.62 ms |

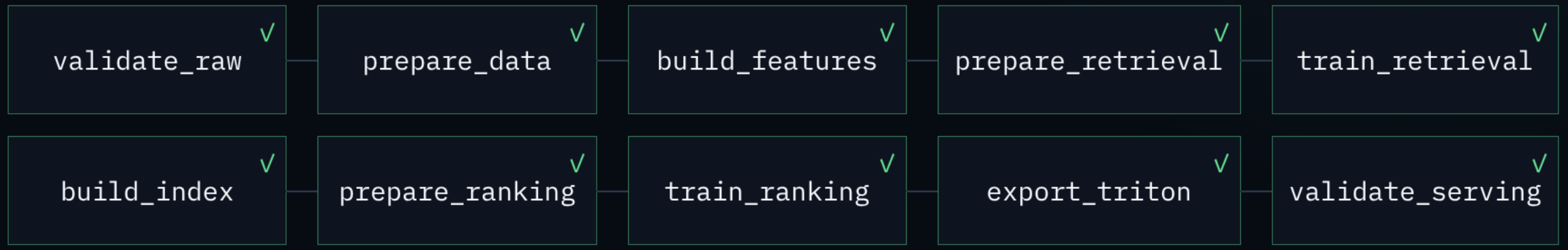
Warm benchmark used the deterministic in-memory TTL fallback. **Redis failures are fail-open**



serving_benchmark.json

ORCHESTRATION

# Airflow turns ten reproducible commands into one validated retraining DAG.




10 / 10 tasks successful · 14 artifact checks · Airflow 3.3.2 · 05 Nov 2026



DAY_10_REPORT.md

EVIDENCE

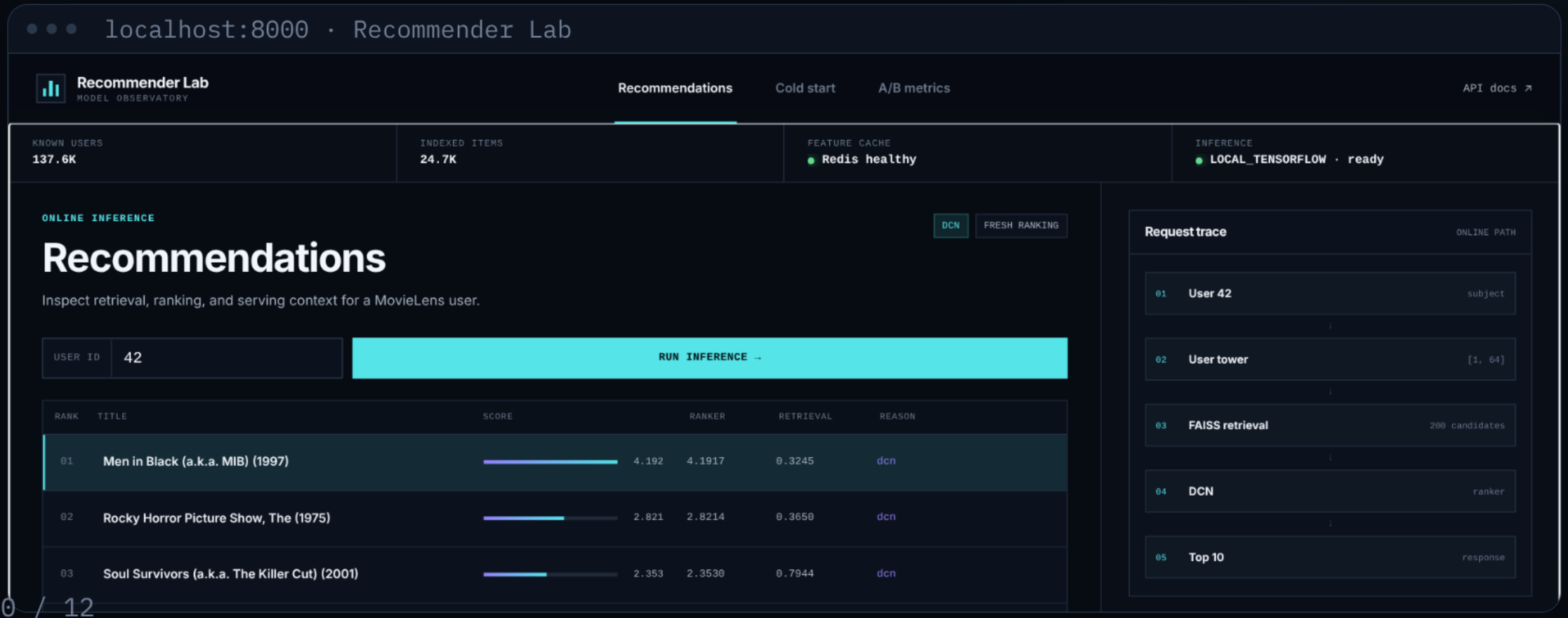
# Core system proof is complete; live Triton remains explicitly optional.

| DATA + FEATURES | RETRIEVAL + INDEX | RANKING |
| --- | --- | --- |
| ✓ Spark integration | ✓ Models + FAISS | ✓ W&D + DCN |
| **SERVING** | **ORCHESTRATION** | **TRITON SERVER** |
| ✓ API + Redis | ✓ Live Airflow DAG | △ Optional live run |

Latest repository validation — **58 passed · 2 service-gated skips**

— LIVE PRODUCT

# The demo exposes the ranking path instead of hiding it.



DEMO RUNBOOK · 4 MINUTES

# One demo proves personalization, fallback, feedback, and observability.

01

### Known user

Run user 42 and inspect the ranked top 10.

02

### Request trace

Show tower → FAISS → DCN → response.

03

### Cold start

Select genres and titles, then generate.

04

### Feedback

Click a result and open A/B metrics.

> Fallback: if a live service fails, use slide 10 and the committed API/Airflow evidence—do not improvise.

— CONCLUSION

# The result is a reproducible recommender system—not a notebook prototype.

01 · REPRODUCIBLE

### Time-safe data and versioned artifacts

Every stage has a deterministic command and an explicit contract.

02 · MEASURED

### Quality and latency trade-offs are visible

Weak results and limitations are reported rather than hidden.

03 · OPERABLE

### Serving, feedback, and orchestration work together

The system can be demonstrated, tested, and extended.